From scalar to vector potential in the magnetostatic solve

Why $\nabla\cdot(\mu\nabla\psi)=0$ fails for non-collinear magnetisation,
the vector-potential replacement, its lattice Boltzmann scheme, and the revised boundary closures

1. The problem

For a non-conducting fluid with no free currents,

$$\nabla\times\boldsymbol{H}=0,\qquad \nabla\cdot\boldsymbol{B}=0,\qquad \boldsymbol{B}=\mu_0(\boldsymbol{H}+\boldsymbol{M}). \tag{1}$$

The magnetisation is *not* taken proportional to the field. It follows the Martsenyuk–Raikher–Shliomis effective-field equation,

$$\boldsymbol{M}=\varphi_p M_s L(\xi_e)\frac{\boldsymbol{\xi}_e}{\xi_e},\qquad \frac{d\boldsymbol{\xi}_e}{dt}=\boldsymbol{\Omega}\times\boldsymbol{\xi}_e-\frac{1}{\tau_B}\frac{L_e}{L_e'}\Big(1-\frac{\boldsymbol{\xi}\cdot\hat{\boldsymbol{n}}_e}{\xi_e}\Big)\hat{\boldsymbol{n}}_e+\frac{1}{2\tau_B}\Big(\frac{\xi_e}{L_e}-1\Big)\big[\boldsymbol{\xi}-\hat{\boldsymbol{n}}_e(\boldsymbol{\xi}\cdot\hat{\boldsymbol{n}}_e)\big], \tag{2}$$

in which the vorticity term tilts $\boldsymbol{\xi}_e$ away from $\boldsymbol{\xi}$. The physical quantity of interest is the antisymmetric part of the magnetic stress,

$$\boldsymbol{F}_m^{\mathrm{rot}}=\frac{\mu_0}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H}), \tag{3}$$

which vanishes identically when $\boldsymbol{M}\parallel\boldsymbol{H}$. Any formulation that presupposes collinearity therefore removes the effect under study before the calculation begins.

2. Why the scalar potential fails

Two independent obstructions arise, at different levels.

2.1 The closure is inconsistent with the magnetisation model

Setting $\boldsymbol{H}=-\nabla\psi$ satisfies $\nabla\times\boldsymbol{H}=0$ automatically and, with $\boldsymbol{M}=\chi\boldsymbol{H}$, gives the familiar variable-coefficient Laplace problem

$$\nabla\cdot(\mu\nabla\psi)=0,\qquad \mu=\mu_0(1+\chi). \tag{4}$$

Equation (4) *assumes* $\boldsymbol{M}=\chi\boldsymbol{H}$, i.e. $\boldsymbol{M}\times\boldsymbol{H}=0$. That is exactly the quantity (3) needs. The susceptibility closure and the MRSh dynamics (2) cannot both hold: adopting (4) sets $\boldsymbol{F}_m^{\mathrm{rot}}\equiv 0$ and the rotational viscosity vanishes identically.

Retaining a general $\boldsymbol{M}$ but keeping the scalar potential is possible in principle. Substituting $\boldsymbol{H}=-\nabla\psi$ into $\nabla\cdot\boldsymbol{B}=0$ gives

$$\nabla^2\psi=\nabla\cdot\boldsymbol{M}, \tag{5}$$

a constant-coefficient Poisson equation whose source is the bound magnetic charge. This is consistent with (2). It fails for a second, geometric reason.

2.2 A periodic scalar potential cannot carry a mean field

Statement. *Let the domain be periodic in a direction $\hat{\boldsymbol{e}}$ with period ℓ, and let ψ be single-valued and periodic. Then $\langle\boldsymbol{H}\cdot\hat{\boldsymbol{e}}\rangle=0$.*

Proof. Let C be any straight path winding once through the domain along $\hat{\boldsymbol{e}}$. Since $\boldsymbol{H}=-\nabla\psi$,

$$\int_C\boldsymbol{H}\cdot d\boldsymbol{\ell}=-\big[\psi\big]_{\text{start}}^{\text{end}}=-\big[\psi(\boldsymbol{x}_0+\ell\hat{\boldsymbol{e}})-\psi(\boldsymbol{x}_0)\big]=0 \tag{6}$$

by periodicity. The left side equals $\ell\,\langle\boldsymbol{H}\cdot\hat{\boldsymbol{e}}\rangle_C$, so the line average of $\boldsymbol{H}\cdot\hat{\boldsymbol{e}}$ vanishes on every such path, and hence over the domain. □

The consequence is direct. The target configuration is a channel, periodic along the flow direction $\hat{\boldsymbol{x}}$, driven by a uniform applied field $\boldsymbol{H}_0 = H_{0x}(t)\hat{\boldsymbol{x}}$ parallel to the flow. By (6) no single-valued periodic ψ can represent it: the mean streamwise field is forced to zero regardless of the boundary data or the source term. The obstruction is topological, not numerical, and no refinement, relaxation setting or boundary treatment removes it.

The usual workaround — adding the applied field analytically, $\boldsymbol{H} = \boldsymbol{H}_0 - \nabla\psi$, and solving for the induced part only — is available, but it places the applied field in the interior as a prescribed term rather than letting it enter through the boundary, which is the opposite of the intended formulation.

3. The vector-potential replacement

Enforce $\nabla \cdot \boldsymbol{B} = 0$ identically instead. In two dimensions with $\boldsymbol{B} = \nabla \times (A\hat{\boldsymbol{z}})$,

$$B_x = \frac{\partial A}{\partial y}, \qquad B_y = -\frac{\partial A}{\partial x}, \qquad \boldsymbol{H} = \frac{\boldsymbol{B}}{\mu_0} - \boldsymbol{M}, \tag{7}$$

and the remaining equation $\nabla \times \boldsymbol{H} = 0$ becomes

$$\boxed{\nabla^2 A = -\mu_0 \left(\nabla \times \boldsymbol{M}\right)_z = -\mu_0 \left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y}\right).} \tag{8}$$

The roles are interchanged relative to (5): the source is the bound *current* $\nabla \times \boldsymbol{M}$ rather than the bound charge $\nabla \cdot \boldsymbol{M}$, and no collinearity is assumed anywhere.

Why the obstruction does not recur. A uniform applied field $H_{0x}\hat{\boldsymbol{x}}$ corresponds to

$$A_0 = \mu_0 H_{0x}\, y. \tag{9}$$

This is linear in the *wall-normal* coordinate, which is bounded, and periodic in x — the direction along which the field points. The representation therefore costs nothing: A remains single-valued and periodic in x while $\langle B_x \rangle = \mu_0 H_{0x} \neq 0$. The complementary structure of (6) is the reason. For ψ the field is minus the gradient *along* the field direction, so a mean field requires a secular potential in a periodic direction. For A the field is the gradient *perpendicular* to the field direction, so the secular growth is pushed into the bounded direction, where it is admissible.

Consequence for the boundary data. Since $\nabla A = \hat{\boldsymbol{z}} \times \boldsymbol{B}$, the normal derivative of A is the *tangential* induction. The applied field can therefore be imposed entirely through the wall condition, with no interior source term — which is the formulation adopted here.

4. Lattice Boltzmann scheme for the potential

Equation (8) is elliptic and is solved as the steady state of a diffusion equation in a pseudo-time,

$$\frac{\partial A}{\partial t} = \varepsilon \nabla^2 A + S, \qquad S = \varepsilon\, \mu_0 \left(\nabla \times \boldsymbol{M}\right)_z. \tag{10}$$

Setting $\partial_t A = 0$ returns (8) for *any* ε, because the same factor multiplies both terms; ε is a solver parameter of dimension $\mathrm{L}^2\mathrm{T}^{-1}$, not a material property, and in particular contains no permeability. The D2Q9 MRT scheme is

$$h_k(\boldsymbol{x} + \boldsymbol{e}_k \Delta t, t + \Delta t) = h_k(\boldsymbol{x}, t) - \left(\mathsf{M}^{-1}\mathsf{S}^h\mathsf{M}\right)_{k\ell}\left[h_\ell - h_\ell^{\mathrm{eq}}\right](\boldsymbol{x}, t) + \Delta t\, c_\Phi\, w_k S, \qquad h_k^{\mathrm{eq}} = w_k A. \tag{11}$$

4.1 Recovered equation

Only two contractions of the collision matrix are needed. Row 0 of M is unity and rows $3, 5$ are e_{kx}/c, e_{ky}/c, so

$$\sum_k \Lambda_{k\ell} = s_0^h, \qquad \sum_k e_{k\alpha}\Lambda_{k\ell} = s_1^h e_{\ell\alpha}, \qquad s_1^h \equiv s_3^h = s_5^h. \tag{12}$$

All other relaxation rates act on ghost moments and do not appear. With $A = \sum_k h_k + aS$, a Chapman–Enskog expansion of (11) gives

$$\frac{\partial A}{\partial t} = \underbrace{\Delta t\, c_s^2 \left(\frac{1}{s_1^h} - \frac{1}{2}\right)}_{\varepsilon} \nabla^2 A + \underbrace{\left(\frac{s_0^h a}{\Delta t} + c_\Phi\right)}_{\mathscr{C}} S + \underbrace{\left[a\left(1 - \frac{s_0^h}{2}\right) - \frac{\Delta t}{2} c_\Phi\right]}_{\mathscr{T}} \frac{\partial S}{\partial t}. \tag{13}$$

Matching to (10) requires

$$\varepsilon = \Delta t\, c_s^2 \left(\tau_h - \tfrac{1}{2}\right), \quad \tau_h = 1/s_1^h; \qquad \mathscr{C} = 1 \ \text{(mandatory)}; \qquad \mathscr{T} = 0 \ \text{(optional)}. \tag{14}$$

The choice $(c_\Phi, a) = (1, 0)$ satisfies $\mathscr{C} = 1$ and is adopted. It leaves $\mathscr{T} = -\Delta t/2$, whose effect is negligible whenever $\omega \ll \varepsilon k^2$ — the same condition that makes the solve quasi-static. *A pair with $\mathscr{C} \neq 1$ rescales every computed field by that factor*, so this normalisation is not cosmetic. The relaxation set is

$$\frac{1}{s_3^h} = \frac{1}{s_5^h} = \frac{\varepsilon}{c_s^2 \Delta t} + \frac{1}{2}, \qquad \mathsf{S}^h = \operatorname{diag}\Big(1.0,\ 1.1,\ 1.1,\ s_3^h,\ s_3^h,\ s_3^h,\ s_3^h,\ 1.2,\ 1.2\Big). \tag{15}$$

4.2 Field recovery

Because $h_k^{\mathrm{eq}} = w_k A$ carries no first moment, $\sum_k \boldsymbol{e}_k h_k^{\mathrm{eq}} = 0$, the first moment of h is purely non-equilibrium and returns ∇A directly. Applying the rotation (7),

$$A = \sum_k h_k, \qquad B_x = -\frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{ky} h_k, \qquad B_y = +\frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{kx} h_k, \qquad \boldsymbol{H} = \frac{\boldsymbol{B}}{\mu_0} - \boldsymbol{M}. \tag{16}$$

No finite-difference stencil is required for the induction, and no additive applied-field term appears: $\boldsymbol{H}$ is obtained from the lattice throughout the interior.

5. Boundary conditions

Let $\hat{\boldsymbol{n}}$ be the outward wall normal, $\hat{\boldsymbol{t}} = \hat{\boldsymbol{n}} \times \hat{\boldsymbol{z}}$, and define

$$\boldsymbol{e}_{\sigma(k)} = \boldsymbol{e}_k - 2(\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}})\hat{\boldsymbol{n}} \ \text{(specular)}, \qquad K_{\mathrm{in}} = \{k : \boldsymbol{e}_k \cdot \hat{\boldsymbol{n}} < 0\}, \qquad W_{\mathrm{in}} = \sum_{k \in K_{\mathrm{in}}} w_k = \frac{c_s^2}{2} = \frac{1}{6}, \tag{17}$$

the last identity following from $\sum_k w_k (\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}})^2 = c_s^2$ for a flat wall, the summand being unity on K_{in} and its mirror image and zero otherwise.

5.1 What changed

	scalar potential ψ	vector potential A
governing equation	$\nabla^2 \psi = \nabla \cdot \boldsymbol{M}$	$\nabla^2 A = -\mu_0 (\nabla \times \boldsymbol{M})_z$
source	bound charge	bound current
wall datum	*normal* component of $\boldsymbol{B}$	*tangential* component of $\boldsymbol{B}$
condition at wall	$\partial\psi/\partial n$ from $\hat{\boldsymbol{n}}\cdot\boldsymbol{B}$ continuity	$\partial A/\partial n = \hat{\boldsymbol{t}}\cdot\boldsymbol{B}$ from $\hat{\boldsymbol{t}}\cdot\boldsymbol{H}$ continuity
applied field enters	interior, additively	wall condition only
mean field along periodic direction	forced to zero	unrestricted

5.2 Derivation of the wall closure

Specular reflection alone transfers nothing, since σ is a weight-preserving bijection $K_{\mathrm{in}} \leftrightarrow K_{\mathrm{out}}$:

$$\sum_{k \in K_{\mathrm{in}}} h_k(t + \Delta t) - \sum_{k \in K_{\mathrm{out}}} h_k^*(t) = 0 \qquad \Longrightarrow \qquad \frac{\partial A}{\partial n} = 0, \tag{18}$$

which is a homogeneous Neumann condition, not the one required. Adding a weighted increment on the incoming links,

$$h_k(\boldsymbol{x}_b, t+\Delta t) = h^*_{\sigma(k)}(\boldsymbol{x}_b, t) + B_w\, w_k, \qquad k \in K_{\text{in}}, \tag{19}$$

transfers $B_w W_{\text{in}}$ per node per step. Equating this to the Fickian flux $\varepsilon\, \partial A/\partial n$ over Δt fixes the coefficient:

$$\boxed{B_w = \frac{\Delta t\, \varepsilon\, \gamma}{W_{\text{in}}} = \frac{2\Delta t\, \varepsilon\, \gamma}{c_s^2} = 6\,\Delta t\, \varepsilon\, \gamma \ \text{(D2Q9)}, \qquad \gamma \equiv \left.\frac{\partial A}{\partial n}\right|_{\text{wall}}.} \tag{20}$$

5.3 The wall datum

From $\nabla A = \hat{\boldsymbol{z}} \times \boldsymbol{B}$ the normal derivative of A is the tangential induction, and continuity of the tangential field across the wall, $\hat{\boldsymbol{t}} \cdot \boldsymbol{H} = \hat{\boldsymbol{t}} \cdot \boldsymbol{H}_0$, gives

$$\gamma = \hat{\boldsymbol{t}} \cdot \boldsymbol{B} = \mu_0 \left(\hat{\boldsymbol{t}} \cdot \boldsymbol{H}_0 + \hat{\boldsymbol{t}} \cdot \boldsymbol{M} \right). \tag{21}$$

For $\boldsymbol{H}_0 = H_{0x}(t)\hat{\boldsymbol{x}}$ and walls normal to $\hat{\boldsymbol{y}}$,

$$\gamma^{\text{bot}} = -\mu_0 (H_{0x}(t) + M_x^w), \qquad \gamma^{\text{top}} = +\mu_0 (H_{0x}(t) + M_x^w), \qquad M_x^w = \tfrac{3}{2} M_x\big|_1 - \tfrac{1}{2} M_x\big|_2, \tag{22}$$

the second-order extrapolation placing γ at the wall rather than at the first fluid node. *This is the only place the applied field appears.* In the x direction the populations stream periodically, unchanged. Where A itself is prescribed rather than its gradient, the anti-bounce-back form $h_k = -h^*_{\bar{k}} + 2 w_k A_w$ applies, second-order accurate at a halfway wall.

5.4 Solvability

Summing (11) over k with (12) and $\mathscr{C} = 1$, the lattice content changes by $\Delta t\, S$ per interior node and by $\Delta t\, \varepsilon\gamma$ per wall node. A steady state exists if and only if

$$-\mu_0 \sum_{\boldsymbol{x}} (\nabla \times \boldsymbol{M})_z = \sum_{\text{walls}} \gamma, \tag{23}$$

which is the discrete divergence theorem, and is the discrete counterpart of the compatibility condition of the pure Neumann problem. With (22) it holds identically: the applied-field contributions cancel between the two walls, being equal and opposite. Residual mismatch between the nine-point gradient stencil and the wall extrapolation is removed by projecting the source onto the range of the discrete Laplacian,

$$\Delta_A = -\mu_0 \sum_{\boldsymbol{x}} (\nabla \times \boldsymbol{M})_z - \sum_{\text{walls}} \gamma, \qquad (\nabla \times \boldsymbol{M})_z \;\leftarrow\; (\nabla \times \boldsymbol{M})_z - \frac{\Delta_A}{N}, \tag{24}$$

with N the number of nodes. In practice Δ_A remains at machine precision, the projection acting only as a safeguard. Because the problem is pure Neumann, A is determined only up to an additive constant; this is immaterial, since (11) is linear in h and every physical quantity depends on A through its gradient alone.